



curved blade adapted for harvesting grain

Reaping hook Date unknown

By the Iron Age, metal harvesting sickles had replaced flint-bladed ones since the metal was readily available and easier to sharpen and mend than flint.



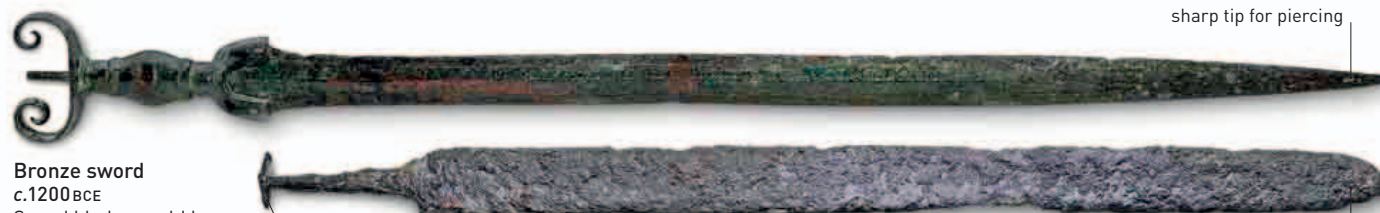
Metal shears Date unknown

These iron shears from Italy are similar to those used by later sheep-shearers. They were found in Riva del Garda, in the Italian province of Trento.



Cast-iron mold c.300 BCE

The Chinese had invented high-temperature furnaces capable of melting iron as early as 500 BCE. This enabled them to produce cast iron by pouring molten metal into molds such as this one, used to make agricultural tools.



Bronze sword c.1200 BCE

Sword blades could be created using bronze, an alloy of copper and tin. Bronze Age swords, such as this one from France, were carried only by the rich.



Iron sword c.500–700

The Anglo-Saxons used the pattern-welding technique to make swords, in which rods of iron were twisted together and forged to form the core. An edge was then added.

flat pommel

sharp tip for piercing

blade with rounded tip

EARLY METALLURGY

ANCIENT METALLURGISTS PRODUCED A VARIETY OF OBJECTS—FROM LETHAL WEAPONRY TO STUNNING JEWELRY

The development of metallurgy, from around 6500 BCE, made possible the production of ornamental objects of great beauty as well as tools and weapons that were more durable and effective than those made of wood.

The earliest metalworking was cold-hammering—the beating of naturally occurring metals. After smelting (heating ore to extract metal) was developed, techniques became more sophisticated. Metal casting began around 5000 BCE, and alloys were developed in the 5th millennium BCE. By the end of the ancient period, techniques such as gilding and inlaying had been developed and metalworking had spread across much of the world.



Chariot decoration c.100 BCE–100 CE

Enameling, or the fusing of molten glass with metal, was invented around 1200 BCE. The use of red glass in enameling became especially popular in the late Iron Age, as seen in this Celtic chariot decoration.



Bronze Celtic brooch c.800 BCE

This ornate brooch was created by Hallstatt craftsmen in Austria. The spiral pattern was part of the Celtic artistic repertoire for over 1,500 years.



bird's head in profile

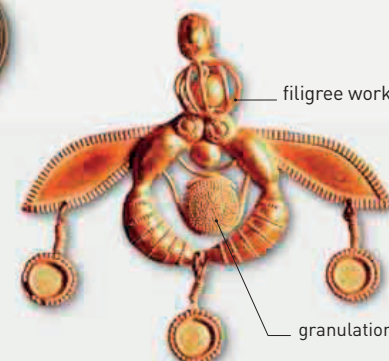
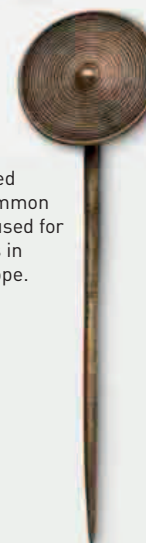
writhing snake pattern

Anglo-Saxon belt buckle c.620

This gold belt buckle features an intertwined pattern of snakes and beasts, highlighted in black niello—an enamel-like substance formed from an alloy of silver, copper, lead, and sulfur.

Bronze pin c.1200

Pins with flattened heads were a common decorative item used for fastening clothes in Bronze-Age Europe.



Gold Minoan pendant c.1700–1550 BCE

This pendant, depicting bees depositing honey on a honeycomb, exhibits granulation (minute balls of gold soldered onto the surface) and filigree (fine threads of metal).